Projects

Start Assignment

- Due Oct 30 by 11:59pm
- Points 5
- Submitting a file upload

General Information

Team size: Three students per group
Project type: Design, implementation, evaluation, and demonstration.

Each group selects one project. The purpose is to build and evaluate a complete computer-vision pipeline, not only to run a pretrained model. Existing libraries and pretrained models can be used with clarity on their source and role in the project.

Common Requirements for All Projects

Each team must:

1. Implement a complete and reproducible pipeline.
2. Establish one simple baseline, as defined below, and compare the final system against it using the same evaluation data and primary metric.
3. Report exact metric definitions, sample counts, parameter settings, software versions.
4. Maintain a short contribution record identifying each member's main responsibilities. Every member must understand the complete system and may be questioned about any component.

Clarification on "establish one simple baseline":

A baseline is a deliberately simple reference solution used to show whether the final design provides a meaningful improvement. It is **not a second complete project** and does not need extensive optimization. The baseline should normally be **implementable using standard library functions** or by **disabling one advanced part of the final pipeline**.
Students should spend only a small part of the project effort on it. They must run the baseline and final system on the same evaluation data and compare them using the project's principal metric. One small results table and a short explanation are sufficient.

Suggested Baseline Examples

Each group should select one simple baseline for comparison with its final system. Equivalent baselines may be used if justified.

Project	Possible baseline	Main comparison
Project 1: Object Detection and Localization	Detector without decoy filtering: treat every detection as real and estimate position using median depth	Real-object TPR, decoy FPR
	Classical red-color and shape segmentation with simple depth estimation	Same metrics
Project 2: 3D Reconstruction and Volume Estimation	Approximate the tank using a simple cylinder or box model	Volume error and runtime
Project 3: Anchor-Based Object Recognition and Localization	Global color histogram comparison (e.g. with cosine similarity)	Precision (Of all frames classified as target-present, how many actually contain the target?)
	ORB or SIFT matching ranked by match count	Same metrics

The baseline and final system must use the same evaluation data. One compact results table and a short explanation of the observed improvement or trade-off are sufficient.

Interpretation of Performance Metrics

The per-project values are course-level performance targets, not automatic pass/fail boundaries. Each project has only one or two principal targets. Other measurements are supporting evidence that helps explain the system. A technically sound system that falls below a target may still pass if the team diagnoses the limitations convincingly.

Project #1: Object Detection and Localization

Objective:
Build a system that detects specific objects (e.g., fire extinguishers) in real-time, while ignoring printed replicas (decoys). Your system should also estimate the 3D position of the real objects.

Requirements:

- Implement or adapt an object-detection model, such as YOLO or Faster R-CNN, using the provided training data.
- Detect and localize fire extinguishers using bounding boxes.
- Design a filtering or classification mechanism that distinguishes real fire extinguishers from printed or displayed replicas.
- Estimate the 3D coordinates of accepted real objects using stereo images, depth measurements, or another suitable sensor-based method.
- Clearly define the coordinate system and units used for the reported 3D position.

- Establish a simple baseline and compare it with the final system using the same evaluation data and metrics.

Resources provided:

- Fire extinguisher dataset for training accessed through this **link** (https://universe.roboflow.com/fire-extinguisher/fireextinguisher-z5atr) .
- Custom dataset with real and decoy objects, access through this **link** (https://ltuse-my.sharepoint.com/:f:/g/personal/christoforos_kanellakis_ltu_se/IgDOCPzeqg2_Rb7kvrlQsd9PARL_-UwkNcTuMzHSEoUbb7Q?e=C7VQ2L) .

Performance metrics

Filtering Accuracy (Decoy Rejection)

Metric	Target Value	Notes
True Positive Rate for real objects	≥ 0.75	Number of real objects detected and accepted as real, divided by the total number of real objects
False Positive Rate for decoys	≤ 0.15	Number of decoys detected and accepted as real, divided by the total number of decoys

For evaluation:

- A true positive for a real object occurs when an annotated real fire extinguisher is correctly detected, accepted as real, and associated with the correct object region.
- A false negative occurs when an annotated real fire extinguisher is missed or incorrectly rejected as a decoy.
- A decoy false positive occurs when an annotated printed or displayed replica is accepted as a real fire extinguisher.
- A detection is associated with an annotated object when the center of the predicted bounding box lies inside the reference bounding box.

Topics to include in the Presentation Slide Deck (Report):

System Architecture & Design

Reflect on

- Training-data preparation.
- Object-detection model.
- Training or model-adaptation procedure.
- Real-versus-decoy filtering mechanism.
- Confidence thresholds and decision rules.
- Depth or stereo processing.
- Conversion from image detections to 3D coordinates.

- Final detection, real/decoy decision, and 3D output.

Evaluation Setup

Describe:

- The training, development, and evaluation subsets.
- The number of evaluated real-object and decoy occurrences.
- How reference bounding boxes and real/decoy labels were obtained.
- The coordinate system and units used for 3D evaluation.
- The detector confidence threshold and decoy-rejection threshold.

Results

Compare the baseline and final systems using:

- Real-object true-positive rate.
- Decoy false-positive rate.
- Raw counts of correctly accepted real objects, missed real objects, and incorrectly accepted decoys.
- 3D localization estimates.

Visual Evidence

Show representative examples of:

- Correct detection and acceptance of a real object.
- Correct rejection of a printed or displayed decoy.
- A missed or incorrectly rejected real object (if it happens).
- A decoy incorrectly accepted as real (if it happens)..
- The estimated 3D coordinates for selected detections.

Reflections and Limitation

Reflect on:

- How the amount and diversity of training data affect generalization.
- How confidence and filtering thresholds affect missed detections and accepted decoys.
- Missing, noisy, or mixed depth measurements near object boundaries.
- How the selected point or depth statistic affects the reported 3D position.
- The trade-off between detection accuracy, decoy rejection, 3D accuracy, and processing speed.

Project #2: 3D Reconstruction and Volume Estimation

Objective:
Build a pipeline to reconstruct objects in 3D from stereo camera data, segment them from the background, and estimate their volume.

Requirements:

- Develop a stereo-vision pipeline that generates a 3D point cloud or mesh from the provided recordings. Existing reconstruction libraries may be used if their role is explained..
- Segment the object using semantic segmentation, edge detection, or instance segmentation techniques.
- Estimate the volume from the segmented 3D model.
- Establish a simple baseline and compare it with the final system using the same evaluation data and volume definition.

Resources provided:

- Custom recordings of various objects, accessed through this **link (https://ltuse-my.sharepoint.com/:f:/g/personal/christoforos_kanellakis_ltu_se/IgDPQPXFjSL7SIt-sxcpPSpJAe_-LtJq1gujdfgCSfTAeUI?e=7QfJ0f)** .

Performance metrics

Segmentation accuracy

Metric	Target Value	Notes
Absolute percentage volume error	+-30% of ground truth	Absolute difference between the estimated and reference volume, divided by the reference volume.

Ground truth volume: Usually the tank-only volue ranges from 8-12 Liters. For the project we define the ground-truth at **~10L**.

Topics to include in the Presentation Slide Deck (Report):

System Architecture & Design

Reflect on

- Stereo input and camera calibration.
- Disparity or depth estimation.
- Invalid-depth handling and depth filtering.
- Conversion from disparity or depth to 3D coordinates.
- Object segmentation.
- Point-cloud cleaning, alignment, or fusion.
- Volume-estimation method.
- Main parameters, coordinate systems, scales, and measurement units.

Evaluation Setup

Describe:

- The calibration information used by the pipeline.
- How frames were selected for development and final evaluation.

- Which parts of the tank were included in the volume definition.
- The units and coordinate system used.

Results

Compare the baseline and final systems using:

- Absolute percentage volume error.
- Processing time.
- Supporting segmentation or valid-depth evidence.
- Variation between estimates if multiple frames or reconstructions were used.

Visual Evidence

Show representative examples of:

- The estimated disparity or depth map.
- The segmented region.
- The raw and cleaned 3D point cloud.
- The final point cloud, surface, mesh, or fitted geometric representation used for volume calculation.
- A visualization clarifying which parts were included in the final volume estimate.

Reflections and Limitations

Reflect on:

- The effect of calibration and rectification errors.
- Depth uncertainty and discontinuities near object boundaries.
- Segmentation errors and background points.
- How depth and segmentation errors propagate into the final volume estimate.
- The most important improvement that could be made with additional views, better calibration, or more reliable depth information.

Project #3: Anchor-Based Object Recognition and Localization

Objective:
Develop a system that uses three provided anchor images of a specific physical object to determine whether that same object is present in each video frame. When the target is present, the system should localize it in the image. Unlike category-level object detection (like in Project#1), the target in this project is defined by the anchor images rather than by a fixed class label or a large labelled training dataset.

Requirements:

- Extract suitable visual features from the provided anchor images and dataset frames.
- Identify or generate image regions that may contain the target object.

- Compare potential target regions with the anchor images using suitable features, descriptors, embeddings, similarity measures, or feature correspondences.
- Assign matching evidence or a similarity score to each candidate.
- Determine whether the target is present using a justified decision rule or threshold.
- When the target is accepted, localize it using a bounding box, polygon, or another suitable region representation.
- Establish a simple baseline and compare it with the final system using the same evaluation frames and metrics.

Resources provided:

- Custom dataset with visually similar objects, accessed through this **link (https://ltuse-my.sharepoint.com/:f:/g/personal/christoforos_kanellakis_ltu_se/IgDRwyr88X-9SavC8-QO8DM1AUNX1Z5KSgED_F8YN72Rpls?e=sqxJcl)** .
- Three anchor images showing the target object from different viewpoints.

Performance metrics

Metric	Target Value	Notes
Presicion	≥ 75%	Of all frames classified as target-present, how many actually contain the target?
Recall	≥ 60%	Of all evaluation frames containing the real target, in how many is it correctly detected and localized?
Localization success	Report	Percentage and number of accepted detections that satisfy the localization criterion.

For evaluation:

- A true positive occurs when the target is present and the system accepts a correctly localized target region.
- A false positive occurs when the system accepts a printed replica, another object, or a background region as the target.
- A false negative occurs when the real target is present but is not correctly detected and localized.
- A localization is considered correct when the center of the predicted region lies inside the reference bounding box.

If the selected method produces several candidate regions, students should rank them according to their matching evidence.

Topics to include in the Presentation Slide Deck (Report):

System Architecture & Design

Reflect on

- Candidate-region generation or localization strategy.
- Feature extraction or image representation.
- Anchor-to-candidate comparison.
- Similarity or matching-score calculation.
- Candidate ranking, where applicable.
- Decision threshold or acceptance rule.
- Final target-present/target-absent decision.
- Final bounding box or localization output.

Evaluation Setup

Describe:

- The number of evaluated frames.
- The number of target-present and target-absent frames.
- How reference bounding boxes or labels were obtained.
- The localization correctness rule.

Results

Compare the baseline and final systems using:

- Precision.
- Recall.
- Raw TP, FP, and FN counts.
- Localization success.

Visual Evidence

Show representative examples of:

- Successful target recognition and localization.
- Correct rejection of printed replicas or visually similar distractors.
- False-positive results.
- False-negative results.
- Candidate regions, feature correspondences, or similarity scores where useful for understanding the decision.

Reflection and Limitation

Reflect on:

- How changes in viewpoint, scale, distance, illumination, and occlusion affect the matching process.
- Whether all three anchor images contribute equally to the result.
- Why printed replicas or background regions may produce strong similarity scores.

- How candidate generation affects the final result.
- How the decision threshold changes the balance between false positives and false negatives.
- The trade-off between accuracy, robustness, and processing time.

Common Project Submission

Submit one ZIP archive containing:

- Code: source files, configuration files, environment specification, and a README with exact reproduction instructions.
- Presentation: the final slide deck in both .pptx and PDF format.
- Demonstration material: a short recorded demonstration or your approach

Presentation Format

- Use approximately 8–12 main presentation slides, including the title and contribution slides but excluding references and the appendix.
- Every team member must present part of the work.
- Prefer diagrams, result tables, plots, and annotated images over dense paragraphs.

Oral Examination Format

The submitted slide deck is used during the oral examination. After the team presentation, each student answers individual questions. The questions may refer to the architecture diagram, a reported result, a failure case, the reflection answers, or another part of the submitted implementation.

Example questions might refer to explain the complete pipeline, interpret one quantitative or qualitative result, predict the effect of changing one parameter.